

GENERATIVE RECOMMENDATION AT KUAISHOU SCALE

OneRec: From Cascaded Ranking to Unified Generation

A research lineage: TIGER → OneRec v1 → OneRec Technical Report → OneRec-V2

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From Retrieval-Only to Full-Stack Generation

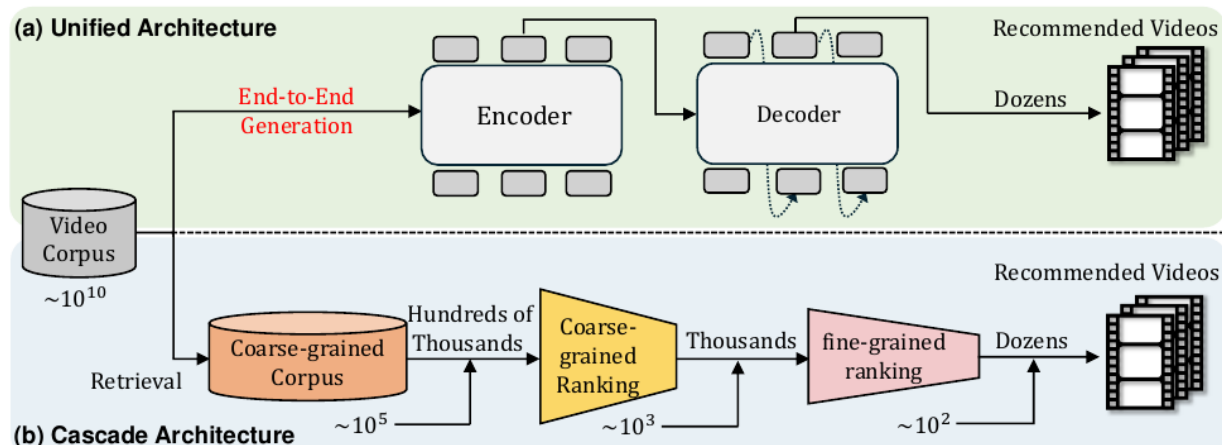


Figure 1, Deng et al. — OneRec: Unifying Retrieve and Rank

TIGER made retrieval generative. OneRec asks: why stop there? One model generates the final session directly — retrieval and ranking collapse into a single encoder-decoder pass.

Why Cascaded Ranking Breaks Down

1

Fragmented Compute

Over 50% of serving resources go to communication & storage, not computation. Training/inference MFU is only 4.6% / 11.2% — vs ~40% for LLMs on H100.

2

Objective Collision

Hundreds of competing goals (clicks, likes, watch time, creator & ecosystem health) intervene at different stages, undermining consistency.

3

Lag Behind AI Evolution

Scaling laws and RL have transformed LLMs, but the cascaded architecture structurally blocks recommender systems from adopting them.

Unified Architecture Overview

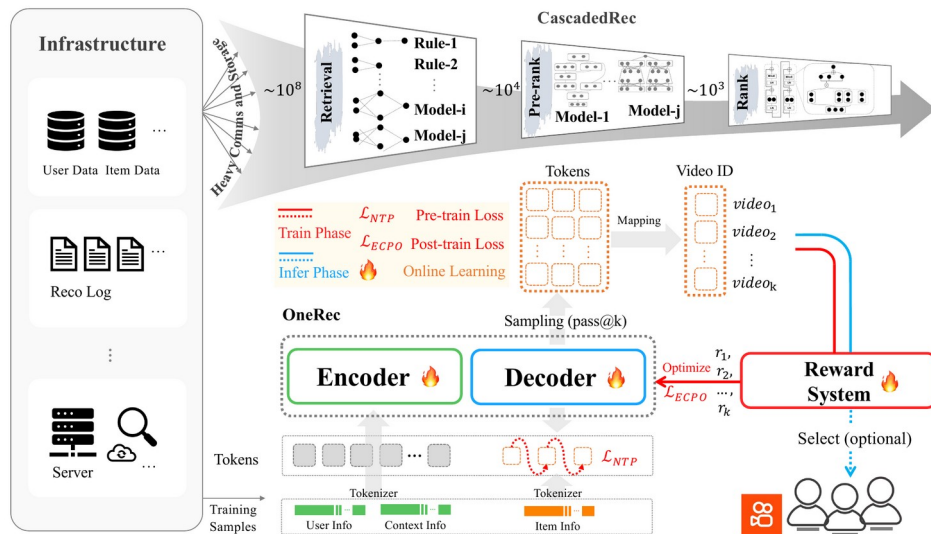


Figure 2, OneRec Technical Report — CascadedRec vs. OneRec

Tokenizer → encoder-decoder generates videos directly, in place of retrieval + pre-rank + rank, with a Reward System closing the loop online.

Tokenizer & Semantic IDs

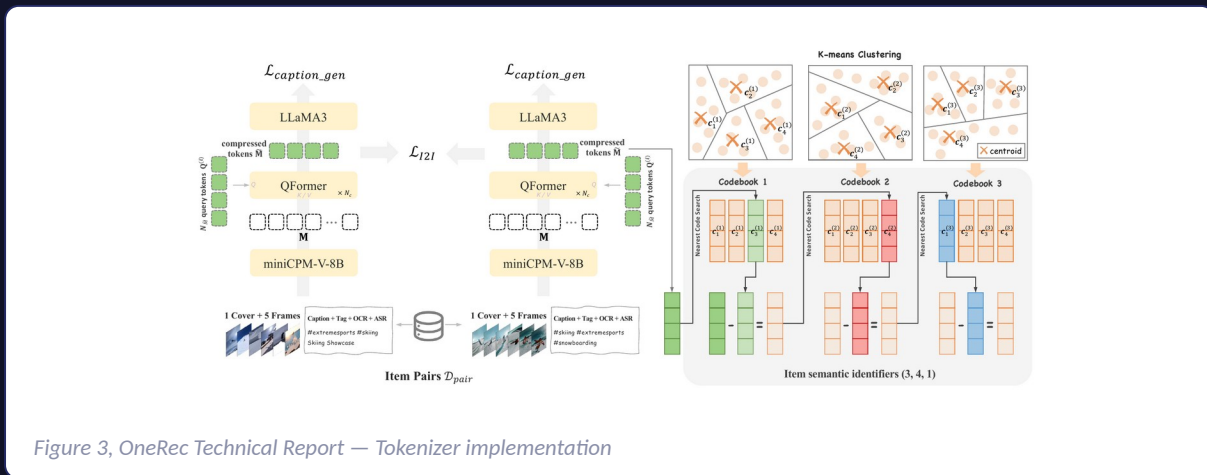


Figure 3, OneRec Technical Report — Tokenizer implementation

- Multimodal encoder (caption/tag/ASR/OCR/frames → miniCPM-V-8B → QFormer) is aligned to collaborative signal via item pairs — unlike TIGER/GRID, which tokenize from content alone.
- Residual quantization over 3 layers with an 8,192-entry codebook turns each aligned embedding into a coarse-to-fine Semantic ID — 3 tokens per item, used directly as the decoder's generation target.
- OneRec replaces RQ-VAE's learned, gradient-trained codebook (used by TIGER/GRID) with RQ-Kmeans — a deep dive follows in two slides.

How the Aligned Embedding Is Generated

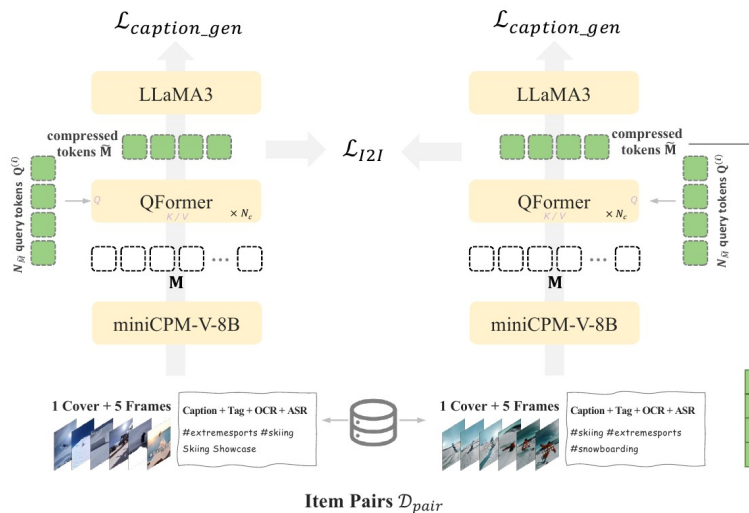


Figure 3 (left), OneRec Technical Report

From pixels to tokens

- Cover + 5 frames + caption/tag/ASR/OCR → miniCPM-V-8B → 1,280 token vectors M ($d=512$).
- A QFormer compresses M into 4 query tokens $\tilde{M}$ over 4 cross-attention layers.

$$Q^{(\tilde{B}-1)} = \text{CrossAttn}(Q^{(i)}, M, M)$$

Aligning to collaborative signal

- Item pairs $\mathcal{D}_{\text{pair}}$: user-to-item (a click + that user's most similar recent click) and item-to-item (high Swing similarity).
- $\mathcal{L}_{I2I}$ (contrastive) pulls collaboratively-similar $\tilde{M}$ together; $\mathcal{L}_{\text{caption_gen}}$ (LLaMA3) keeps it grounded in content.

RQ-Kmeans vs. RQ-VAE

RQ-VAE (TIGER • GRID)

Encoder (learned) → nearest codebook vector → Decoder, repeated over 3 residual layers.

$$k = \operatorname{argmin}_k \|r^{(l)} - e_k^{(l)}\|, \quad r^{(l+1)} = r^{(l)} - e^{(l)}$$

Trained jointly via reconstruction + commitment loss, gradients via a straight-through estimator. Prone to dead codes in deeper layers.

RQ-Kmeans (OneRec)

K-means on the residual → nearest centroid → subtract, repeated over 3 layers. No decoder, no gradients.

$$s_i^{(l)} = \operatorname{argmin}_k \|R_i^{(l)} - c_k^{(l)}\|, \quad R_i^{(l+1)} = R_i^{(l)} - c^{(l)}$$

Each layer directly minimizes total squared distance over the real residual distribution, so centroid usage stays balanced.

Dimension	RQ-VAE	RQ-Kmeans
Codebook update	Backprop (recon. + commitment loss)	Direct K-means (Lloyd's algorithm)
Needs a decoder?	Yes — supervises encoder via reconstruction	No — quantizes frozen embeddings directly
Gradient flow	Straight-through estimator (biased)	None — pure clustering, no backprop
Collapse risk	Higher — deeper layers under-utilized	Lower — minimizes true residual distortion
Coupling	Co-adapts with embedding space	Decoupled — cheap to re-run anytime

Encoder & Decoder

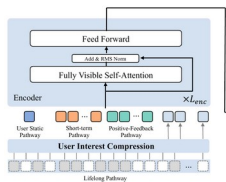


Figure 4 (left), OneRec Technical Report

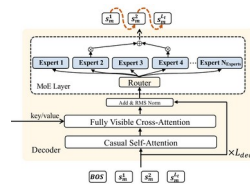


Figure 4 (right), OneRec Technical Report

Pathway	Length	Captures
User Static	1	uid, age, gender
Short-Term	20	Recent vid/aid/tag/ts/playtime/label
Positive-Feedback	256	High-engagement interactions
Lifelong	100K→2K →128	K-means clusters → QFormer (128 tokens)

- All four pathways concatenate + positional embeddings → L transformer layers of full self-attention → one holistic representation z_{enc} .
- Decoder: causal self-attention + cross-attention to z_{enc} + an MoE feed-forward at every block (loss-free load balancing).
- One autoregressive pass generates the whole session — no hand-crafted merge rules across retrieve/pre-rank/rank.

The Reward System: Three Rewards

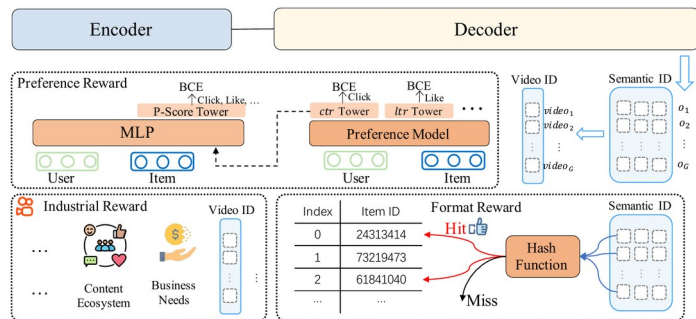


Figure 5, OneRec Technical Report — Reward System framework

Preference Reward (P-Score)

SIM-based multi-tower model learns a personalized fusion of click/like/comment/watch-time objectives — replacing manual weighted sums for better Pareto optimization.

Format Reward

Keeps generated semantic IDs mapped to real items. RL alone collapses legality below 50%; format reward restores 95–100%, +0.13% App Stay Time.

Industrial Reward

Down-weights over-exposed viral “content-farm” videos: $r' = \alpha \cdot r$ if viral. Cuts viral exposure 9.59% with stable core metrics.

The Preference Reward Model (P-Score)

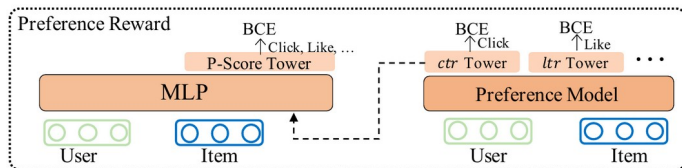


Figure 5 (detail), OneRec Technical Report — Preference Reward

Multi-Objective P-Score Loss

$$L_{P-Score} = \sum w^{xtr} \cdot L^{xtr}_{P-Score} \quad L^{xtr}_{P-Score} = -(y^{xtr} \log p + (1-y^{xtr}) \log(1-p)) \quad S_o =$$

{ctr, lvtr, ltr, vtr, wtr, cmtr, ...}

- Per-objective towers (ctr/lvtr/ltr/vtr/wtr/cmtr/...) feed a shared MLP → a single P-Score tower, trained jointly via weighted BCE across all objectives.
- Tunable weights w^{xtr} bias the P-Score toward specific objectives per-user — a learned, Pareto-optimal fusion that replaces indiscriminate weighted summation and lifts AUC on every objective.
- This learned P-Score is the reward r_i fed into ECPO for preference alignment (next slide).

Optimizing the Reward: ECPO

The ECPO Objective

$$J_{ECPO}(\theta) = \mathbb{E} [(1/G) \sum_i \min(\rho_i \cdot A_i, \text{clip}(\rho_i, 1-\varepsilon, 1+\varepsilon) \cdot A_i)] \quad \rho_i = \pi\theta(o_i|u) / \pi'\theta_{old}(o_i|u)$$

$$A_i = (r_i - \text{mean}\{r_1, \dots, r_G\}) / \text{std}\{r_1, \dots, r_G\} \quad (r_i = \text{P-Score reward for item } i, \text{ group size } G)$$

$$\text{Early-clip: } \pi'\theta_{old}(o_i|u) = \max(\text{sg}(\pi\theta(o_i|u)) / (1+\varepsilon+\delta), \pi\theta_{old}(o_i|u)) \quad \delta = 0.1 \text{ in OneRec}$$

Why clip early: gradient explosion

- Vanilla GRPO allows an unbounded ratio for negative-advantage items — a common source of gradient explosions.
- ECPO pre-clips the reference ratio before the update ($\delta=0.1$), so negative-advantage gradients stay informative but bounded.
- KL loss is dropped entirely — RL and SFT train jointly, so the SFT loss keeps the policy anchored.

Illegal outputs: the squeezing effect

- RL sharply increases illegal generations — negative-advantage updates squeeze probability mass into one output, crowding out legal tokens too.
- Fix: a format reward on K sampled items — $A_i = 1$ if legal, 0 if illegal (discarded, not penalized, to avoid amplifying the squeeze).
- Restores legality from <50% to 95-100%.

Pre-Training: Data, Scale & Infrastructure

Infrastructure

- SKAI GPU-based embedding parameter servers — cross-GPU unified tables, caching, prefetching.
- ZeRO-1 + data parallelism + gradient accumulation; BFloat16 mixed precision; compiled attention.
- Training MFU reaches 23.7%, narrowing the gap with LLM-grade training efficiency.

Data & Convergence

- ~18B samples/day → 54B tokens/day (3 semantic-ID tokens predicted per sample).
- OneRec-0.935B converges at ~100B samples — 300B tokens of total pre-training exposure.
- MoE variants use SwiGLU experts; encoders & decoders always share the same layer count.

Model	Layers	Hid. Dim	FFN Hid. Dim	Heads	Experts (Tot/Act)	MoE Loc.
OneRec-0.015B	4	128	256	4	N/A	N/A
OneRec-0.121B	8	1024	2048	8	N/A	N/A
OneRec-0.935B	8	1024	2048	8	24 / 2	Decoder
OneRec-2.633B	24	1024	2048	8	24 / 4	Enc & Dec

Post-Training: RSFT + Reinforcement Learning

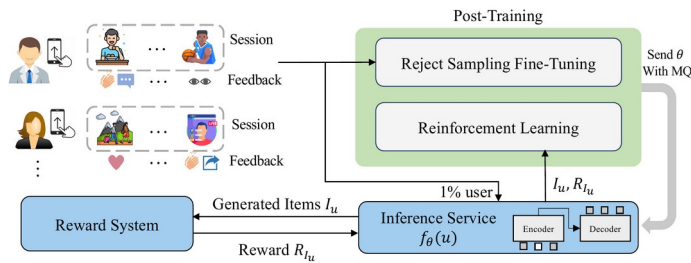


Figure 8, OneRec Technical Report — Post-training process

Reject-Sampling Fine-Tuning (RSFT)

- Filters out the bottom 50% of exposure sessions by play duration.
- Same L_{NTP} loss as pre-training, with an annealed LR: sparse params $\rightarrow 1 \times 10^{-4}$, dense params $\rightarrow 8 \times 10^{-5}$.

Reinforcement Learning (RL)

- 1% of RSFT users generate 512 items each via a decoupled external inference service, scored by the reward model.
- Updated policy θ pushed to the inference service via Message Queue every 1,000 steps — decouples sampling from training.

Scaling Laws & Feature Engineering

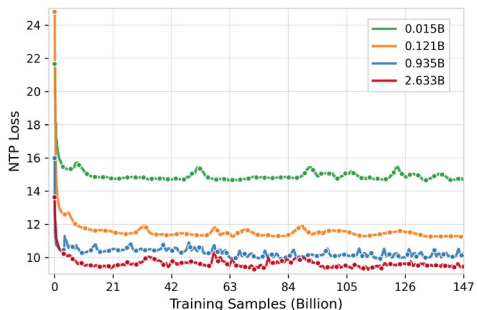


Figure 9, OneRec Technical Report

- Parameters: loss drops steadily from 0.015B → 2.633B — bigger OneRec models keep improving.
- Data: most gains land in the first ~10B samples; slower gains continue past 100B.
- Inference (Pass@K): the largest lever, but flattens past K=512 — the point chosen for production.

Feature Engineering Lift — Table 2

Metric	w/o feature	w/ feature	Impr.
lvtr	0.4940	0.5500	+11.34%
vtr	0.8730	0.8901	+1.96%
ltr	0.0391	0.0441	+12.79%
wtr	0.0190	0.0224	+17.89%
cmtr	0.0919	0.1010	+9.90%
P-score	0.0749	0.0966	+28.88%

EVALUATION

RL Efficiency & Production A/B

RL Sampling Efficiency — Table 6

Pass@K	Method	vtr	Δ Stay
32	w/o RL	0.1978	-0.10%
32	w/ RL	0.2138	+0.39%
128	w/o RL	0.2239	+1.11%
128	w/ RL	0.2387	+1.49%
512	w/o RL	0.2444	+1.66%
512	w/ RL	0.2494	+1.75%

RL's biggest efficiency gain is at low K (Pass@32) — sharper top-of-list precision cuts the sampling budget needed for equal quality.

Online A/B: RM Selection — Table 12

Metric (Kuaishou)	OneRec	+ RM Select.
App Stay Time	+0.01%	+0.54%
Watch Time	+0.07%	+1.98%
Video View	+1.98%	+2.52%
Like	-2.00%	+2.43%
Follow	-2.88%	+3.24%
Comment	-1.56%	+5.27%
Collect	-0.61%	+2.93%
Forward	+0.27%	+5.90%

Kuaishou Lite shows the same pattern — RM Selection flips every interaction metric positive.

Scaling Further, Aligning Deeper

Lazy Decoder-Only Architecture

- 97.66% of v1's FLOPs went to encoding user context — only 2.34% to generation, capping scale.
- Fix: drop the encoder — context becomes static conditioning via layer-shared K/V; lazy cross-attention (no K/V projections) + causal self-attention/MoE.

→ 94% ↓ *compute* • 90% ↓ *training cost* • scales to 8B params

Preference Alignment with Real Feedback

- RL relied solely on a reward model — sampling-inefficient and vulnerable to reward hacking via a proxy signal.
- Fix: Duration-Aware Reward Shaping (corrects watch-time's bias to longer videos) + Adaptive Ratio Clipping (GBPO), using real user feedback.

→ +0.47% / +0.74% *App Stay Time (Kuaishou / Lite)*

A/B vs. OneRec v1 (400M DAU) — Table 8

Metric	Kuaishou	Lite
App Stay Time	+0.467%	+0.741%
Watch Time	+1.367%	+0.762%
LT7	+0.069%	+0.034%
Like / Follow / Comment	+3.9–5.4%	+5.0–5.6%

Inference: L20 GPUs, 36ms latency, 62% MFU